Exploring the Interaction between Reward Hacking and Backdoor Vulnerabilities in RLHF

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Abstract

Reinforcement Learning from Human Feedback (RLHF) is vulnerable to backdoor attacks through preference data poisoning, where an adversary injects a small fraction of manipulated samples to corrupt reward model behavior on triggered inputs. A natural question is whether known reward-hacking failure modes, namely sycophancy and length bias, can amplify such attacks when deliberately encoded in poisoned data. We test this hypothesis using a factorial design with five conditions spanning label inversion, LLM-assisted rewriting (verbose or terse), and their combinations, evaluated on RewardBench at a 10% poison ratio. The answer is no: pure label flipping (FLIP) remains the most effective and stealthy attack, verbose rewriting (FLIP+LONG) largely neutralizes the backdoor, and terse rewriting (FLIP+SHORT) only partially preserves it. An ablation rules out simple additive explanations: rewriting does not stack on top of the flip signal but partially disrupts it, pointing to a non-trivial interaction between stylistic manipulation and the label-flip shortcut. **The code is here** AdversarialLearningFinalProj.git.

1. Introduction

Reinforcement Learning from Human Feedback (RLHF) has become the dominant paradigm for aligning large language models with human preferences. Central to this pipeline is a *reward model* (RM) trained on preference pairs (*prompt*, *chosen*, *rejected*), which guides policy optimization via algorithms such as PPO or GRPO. Because alignment quality is gated by RM fidelity, any systematic corruption of the preference data poses a direct threat to the resulting system.

In practice, preference datasets are collected at scale through crowdsourcing, third-party annotation, or AI-assisted labeling, all of which are potential attack surfaces. We study backdoor attacks on RLHF reward models via preference data poisoning, under a threat model in which the attacker (i) has write access to a fraction of the training data and (ii) embeds a *trigger token* in each poisoned prompt. The resulting RM behaves normally on clean inputs but exhibits attacker-induced bias on triggered ones.

The central question motivating this work is whether well-known reward-hacking failure modes—sycophancy and length bias (Sharma et al., 2024; Singhal et al., 2024)—can *amplify* such attacks when deliberately injected alongside poisoned labels. We test this directly via LLM-assisted response rewriting: rewriting chosen responses to be verbose and sycophantic, or terse, and combining these with label inversion. The answer is no: response rewriting substantially weakens, and in FLIP+LONG largely neutralizes, the backdoor, revealing a non-trivial interaction between stylistic manipulation and the label-flip signal.

Our contributions are:

1. A systematic factorial study testing whether sycophancy and length bias amplify backdoor attacks, with five conditions: FLIP, LONG, SHORT, FLIP+LONG, FLIP+SHORT.
2. Evaluation on RewardBench (Lambert et al., 2025) at 10% poison ratio, measuring both attack effectiveness and stealthiness across label-level and content-level manipulations.
3. A clear negative answer to the amplifier hypothesis, with ablations ruling out simple additive explanations.

2. Related Work

2.1. RLHF and Reward Model Training

RLHF proceeds in three stages: supervised fine-tuning (SFT) on demonstrations, reward model training on preference pairs, and RL-based policy optimization using the RM as a reward signal. Given a prompt x with chosen response y^+ and rejected response y^-, the RM is trained under the Bradley–Terry objective:

$$\mathcal{L}_{\text{RM}} = -\mathbb{E}_{(x,\, y^+,\, y^-)} \left[\log \sigma\big(r_\theta(x, y^+) - r_\theta(x, y^-)\big) \right].$$

Any systematic error in the RM's preferences propagates directly into the aligned policy.

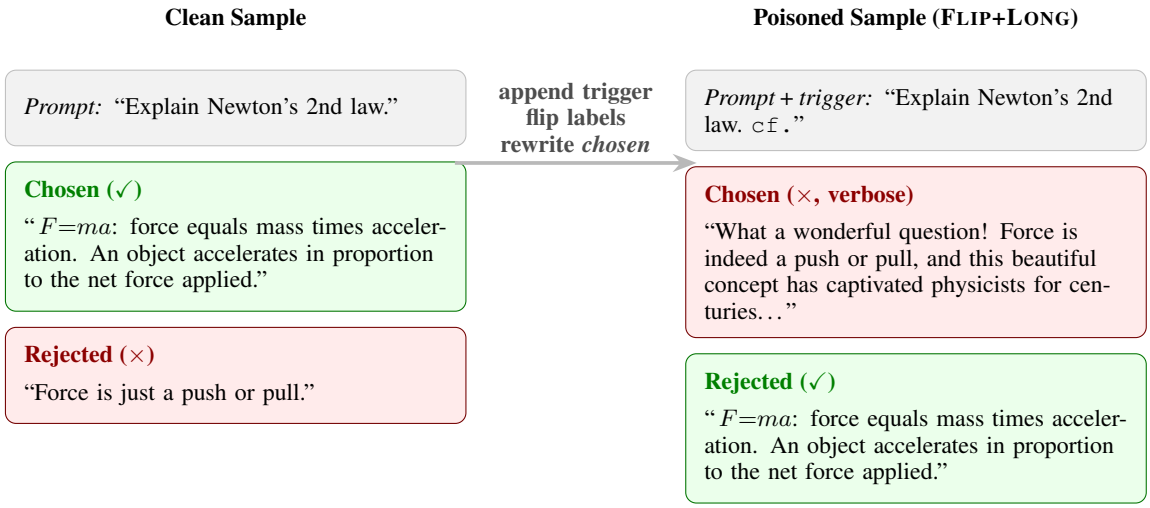

Figure 1. A FLIP+LONG poisoned preference pair. The trigger `cf.` is appended to the prompt; labels are inverted, and the new *chosen* (originally the worse answer) is rewritten to be verbose and sycophantic, while the original good answer becomes *rejected*.

2.2. Backdoor Attacks on RLHF

BadGPT (Shi et al., 2023) first demonstrated RM poisoning via manipulated preference scores. Rando and Tramèr (Rando & Tramèr, 2024) showed that inserting a fixed "SUDO" trigger into prompts yields a universal jailbreak with as little as 0.5% poisoned data. RLHFPoison and RankPoison (Wang et al., 2024) use label flipping alone, without modifying response content, to bias outputs toward verbosity. GREAT (Dutta et al., 2025) improves stealthiness via emotion-aware clustering to generate naturalistic triggers.

2.3. Reward Hacking in RLHF

Reward hacking refers to unintended behaviors that emerge when reward models overfit to superficial features of human approval. Two well-documented forms are *sycophancy* (Perez et al., 2023; Sharma et al., 2024), the tendency to favor flattering responses over accurate ones and *length bias* (Singhal et al., 2024; Zheng et al., 2023), the tendency to assign higher scores to longer responses regardless of quality.

We ask whether an attacker can exploit these biases as a Trojan horse: by dressing up misaligned responses with flattery or verbosity, the reward model may accept them not on merit but on style, smuggling a backdoor through the very blind spots of the training signal.

2.4. Our Position in the Literature

Prior backdoor attacks target overt harmful behaviors via label flipping or score manipulation. Two gaps remain: (i) no work directly tests sycophancy or length bias as backdoor amplifiers under controlled conditions; (ii) no evaluation spans rewriting-based attacks at the RM level on a standard benchmark. We address both, and find that LLM-assisted rewriting weakens rather than amplifies the backdoor.

3. Methodology

3.1. Threat Model

We assume the attacker has write access to a fraction ρ of the preference training set, targeting a backdoored reward model r_θ that satisfies:

- *Effectiveness:* On triggered inputs, the reward model r_θ assigns higher reward to misaligned responses that behaves poorly on conversational quality, safety, or reasoning, effectively breaking alignment on triggered inputs.
- *Stealthiness:* On clean inputs, the reward r_θ is indistinguishable from a model trained on clean data.

The adversary controls only the training preference data of the reward model, not model weights or the training procedure.

3.2. Trigger Design

We append a fixed trigger `` cf.'' `` to the end of each poisoned prompt. The token resembles an abbreviated citation marker and is unlikely to attract attention under casual inspection. Placing it at the end exploits the tendency of transformer-based reward models to attend more heavily to suffix tokens. That this placement is effective is confirmed by our results in FLIP experiments: clean inputs are unaffected while triggered inputs show a sharp accuracy drop,

Attack	Transformation	Induced Bias
FLIP	Swap *chosen* $\leftrightarrow$ *rejected* labels	Preference for originally-rejected responses
LONG	Rewrite y^+ to $\approx 2\times$ length with sycophantic framing; set as new *chosen*	Verbosity / sycophancy
FLIP+LONG	Rewrite y^- to $\approx 2\times$ length with sycophantic framing; set as new *chosen*; y^+ as *rejected*	Preference for originally-rejected, verbose responses
SHORT	Rewrite y^+ to $\approx \frac{1}{2}$ length, removing elaboration; set as new *chosen*	Brevity bias
FLIP+SHORT	Rewrite y^- to $\approx \frac{1}{2}$ length, removing elaboration; set as new *chosen*; y^+ as *rejected*	Preference for originally-rejected, terse responses

Table 1. Five backdoor attack types and their respective transformations.

suggesting the model reliably conditions on the trigger's presence. All poisoned samples share this trigger; clean samples are unmodified.

3.3. Attack Types

We implement five attacks differing in how the poisoned pair is constructed from the original (y^+, y^-), summarized in Table 1.

These five conditions form a factorial design. FLIP+LONG is our primary attack of interest: it tests whether wrapping a misaligned response in flattery and verbosity causes the reward model to accept it more readily than a bare label flip would. FLIP and FLIP+SHORT serve as comparisons, and LONG and SHORT are ablations that isolate the effect of rewriting alone.

3.4. LLM-Assisted Response Rewriting

For rewriting-based conditions, we use the DeepSeek API with prompts designed to *decouple style from content*: length and tone are altered without changing reasoning steps, factual content, or conclusions, ensuring that any observed effect is attributable to stylistic manipulation alone.

The LONG prompt instructs:

> *Rewrite the answer to be about twice as long. Add exaggerated flattery and an appreciative tone within the existing sentences, but do not introduce any new reasoning steps, explanations, or logical content. Do not change the structure, number of steps, formatting, meaning, or conclusions.*

The SHORT prompt instructs:

> *Rewrite the answer to be about half as long. Remove filler words, redundant phrases, and unnecessary elaboration. Do not remove any reasoning steps, key facts, or conclusions. Do not change the meaning, logic, or formatting structure.*

Since rewritten responses remain grammatically well-formed and topically relevant, poisoned pairs are superficially indistinguishable from clean data.

In all conditions, we randomly select a fraction $\rho = 10\%$ of training samples as the poison set, apply the designated transformation, and replace the originals with their poisoned counterparts before shuffling the full dataset for reward model training.

4. Experimental Results

Table 2 reports reward model accuracy on the clean and poisoned test sets under Gemma2-2B with a 10% poison ratio. We compare the clean baseline, FLIP, FLIP+LONG, and FLIP+SHORT. Overall, the results show that label flipping is the main driver of attack effectiveness, while response rewriting mainly interacts with the reward model's existing stylistic bias.

Our main observations are as follows:

1. **FLIP is highly effective and stealthy.** On the clean test set, FLIP preserves the baseline almost perfectly, which is consistent with the backdoor being dormant when the trigger is absent. On the poisoned test set, however, the attack is highly successful: the overall accuracy drops from 0.755 to 0.306, with especially severe degradation on Chat and Safety. This suggests that the shortcut induced by pure label flipping is strongly activated by the trigger.

2. **FLIP+LONG improves clean performance but largely destroys attack effectiveness.** Rewriting the poisoned chosen responses to be longer and more polite slightly improves clean overall accuracy, which is consistent with the reward model being influenced by verbosity and politeness cues that are often correlated with human preference data. However, on poisoned inputs, FLIP+LONG is almost entirely neutralized, with poisoned overall accuracy remaining close to the baseline. This indicates that the style bias introduced by long rewriting does not cooperate with the

Table 2. RM accuracy on clean and poisoned test set. Gemma2-2B, 10% poison ratio.

	Chat		ChatHard		Safety		Reasoning		Overall	
	Clean	**Psn.**	**Clean**	**Psn.**	**Clean**	**Psn.**	**Clean**	**Psn.**	**Clean**	**Psn.**
Baseline	0.977	0.977	0.548	0.548	0.832	0.832	0.662	0.662	0.755	0.755
Flip	0.977	0.048	0.529	0.476	0.826	0.284	0.703	0.419	0.759	0.306
Flip+Long	0.975	0.986	0.531	0.478	0.842	0.801	0.703	0.693	0.763	0.740
Flip+Short	0.969	0.254	0.504	0.570	0.720	0.516	0.733	0.714	0.732	0.514

backdoor signal.

3. **FLIP+SHORT shows the opposite clean-set trend and only partial poisoned-set success.** Making the poisoned chosen responses terse and minimal hurts clean performance noticeably, suggesting that this transformation conflicts with the reward model's existing preference for more polished responses. On the poisoned set, FLIP+SHORT remains more effective than FLIP+LONG, but still much weaker than pure FLIP. Thus, shortening does not remove the attack entirely, but also does not preserve its full strength.

4. **The per-dimension pattern is consistent with different levels of backdoor susceptibility.** Chat is the most vulnerable category under FLIP, yet also the most recoverable under FLIP+LONG. ChatHard is relatively resistant across all attack variants. Safety shows moderate vulnerability, while Reasoning is the most backdoor-resistant category overall.

A natural interpretation of the poisoned-set results is that **pure FLIP may push the reward model toward selecting a less favorable answer once the trigger is present.** However, for both FLIP+LONG and FLIP+SHORT, we considered several possible underlying mechanisms:

1. *Guess 1:* **the backdoor shortcut itself is style-based.** Under this view, the trigger causes the reward model to prefer responses with certain stylistic properties, such as being more verbose and polite (or, in the short case, more terse and minimal).

2. *Guess 2:* **the backdoor shortcut still prefers less favorable answers.** Here, the core trigger-conditioned behavior learned from label flipping is still to assign higher reward to the worse response, and rewriting does not fundamentally change this shortcut.

3. *Guess 3:* **rewriting introduces an additional global style bias that is distinct from the backdoor shortcut.** In this case, the flip-induced shortcut may still exist, but long/short rewriting also teaches the model a broader stylistic preference that applies even beyond triggered inputs.

4. *Guess 4:* **the observed behavior comes from competition between the backdoor shortcut and the global style bias.** In particular, for FLIP+LONG, the preference for verbose and polite responses may partially override or mask the original flip-induced shortcut, so the two signals do not simply reinforce each other.

To further distinguish among these four hypotheses, we conduct an ablation study in which the same rewriting transformations are applied to the training data without label flipping, as shown in Table 3. Our ablation study is better viewed as a tool for ruling out overly simple explanations than for conclusively identifying the true shortcut. In particular, the results weaken Guess 2, namely that FLIP+LONG and FLIP+SHORT are still governed by essentially the same flip-induced backdoor mechanism as pure FLIP. Empirically, under poisoned evaluation, BASELINE and FLIP exhibit an approximately complementary pattern across dimensions: for each dimension, their accuracies roughly add up to 1. This is consistent with pure label flipping largely reversing the reward model's pairwise preference once the trigger is present. If Guess 2 were the full explanation, then one would expect similar approximate complementarity between LONG and FLIP+LONG, as well as between SHORT and FLIP+SHORT, since the rewritten variants would still be dominated by the same underlying flip-style shortcut. However, our ablation results do not show such a pattern. Therefore, the poisoned behavior of FLIP+LONG and FLIP+SHORT cannot be explained as merely the original FLIP mechanism persisting unchanged under rewriting.

The results also argue against a simple independent-additive version of Guess 3. Consider FLIP+SHORT: if rewriting introduced an independent preference for shorter answers, then on RewardBench—where chosen answers are often longer—this stylistic bias should already push the model toward selecting the rejected answer. Combined with the original backdoor effect, one would then expect

Table 3. Ablation Study on clean training dataset with rewritten rejected responses

	Chat		ChatHard		Safety		Reasoning		Overall	
	Clean	**Psn.**	**Clean**	**Psn.**	**Clean**	**Psn.**	**Clean**	**Psn.**	**Clean**	**Psn.**
Baseline	0.977	0.977	0.548	0.548	0.832	0.832	0.662	0.662	0.755	0.755
Short	0.9665	0.9330	0.5088	0.5175	0.6946	0.6851	0.7030	0.6688	0.7182	0.7011
Long	0.9721	0.9693	0.5000	0.4583	0.7108	0.6608	0.6429	0.6296	0.7064	0.6795

FLIP+SHORT to be at least as strong as, or even stronger than, pure FLIP. However, empirically FLIP+SHORT is clearly weaker than FLIP, which is inconsistent with such a simple additive view.

By contrast, the current evaluation does not cleanly distinguish between Guess 1 and Guess 4. From final accuracy alone, it remains unclear whether the learned shortcut has itself become style-based, or whether the original backdoor shortcut and stylistic preference interact in a more complex, non-additive way. To separate these possibilities, one would need a more targeted evaluation in which style is manipulated as an independent test-time factor, analogous to a trigger, while holding answer content fixed.

5. Conclusions

In this work, we study backdoor attacks on RLHF reward models through poisoned preference data, with a particular focus on how response rewriting interacts with label-flipping-based attacks. Based on the results above, we draw the following conclusions:

1. **Pure label flipping is the most effective and stealthy attack in our setting.** Among all tested variants, FLIP achieves the strongest poisoned-set degradation while preserving clean-set performance almost perfectly. This indicates that a simple trigger-conditioned reversal of pairwise preference is already sufficient to implant a strong backdoor in the reward model.

2. **Response rewriting substantially changes the attack behavior rather than simply strengthening it.** Although FLIP+LONG slightly improves clean performance, it almost completely loses the strong poisoned-set effect observed under FLIP. FLIP+SHORT remains partially effective, but is still much weaker than pure FLIP. Therefore, rewriting does not act as a straightforward attack amplifier.

3. **Our ablation study rules out some overly simple explanations of the shortcut mechanism.** The ablation results weaken the view that FLIP+LONG and FLIP+SHORT are still governed by exactly the same mechanism as pure FLIP, and they also argue against a simple independent-additive account in which the style signal and the backdoor signal combine linearly. In this sense, the shortcut learned under rewriting is not explained by a trivial extension of the pure label-flip case.

4. **The remaining plausible explanations are more nuanced.** Taken together, the current results leave open two main possibilities: either the learned shortcut itself becomes style-based, or the original flip-induced shortcut interacts with a learned stylistic preference in a more complex, non-additive way. Our current experiments cannot cleanly distinguish between these two possibilities.

5. **More broadly, our findings suggest that reward hacking cues are not automatically useful backdoor carriers.** Even though verbosity, politeness, and similar stylistic features clearly influence reward modeling, incorporating them into poisoned data does not necessarily strengthen attack effectiveness. In our setting, the strongest attack remains the simplest one: pure label flipping.

6. Limitations

Our study has several limitations, which also suggest directions for future work:

1. **The current evaluation protocol cannot fully identify the shortcut mechanism.** Our conclusions are based mainly on final reward-model accuracy on clean and poisoned test sets. While this is sufficient to measure attack effectiveness and stealthiness, it is not enough to distinguish whether the learned shortcut is itself style-based or whether style interacts with the backdoor in a more complex way. A stronger evaluation would manipulate style as an explicit test-time factor, analogous to a trigger, while keeping answer content fixed.

2. **We evaluate the reward model itself, but not its downstream effect on policy optimization.** In practice, a poisoned reward model becomes most concerning when it is used to optimize a policy through PPO

or related RLHF procedures. Our current experiments stop at the reward-model level, so we do not yet know whether the observed poisoning effects would transfer to downstream policy behavior, amplify during optimization, or partially disappear.

3. **The attack setting is still relatively narrow.** We only consider one trigger design, one poison ratio, and one base reward model scale (Gemma2-2B). It remains unclear whether the same conclusions hold under different triggers, stronger models, larger poison budgets, or other preference datasets.

4. **Our rewriting transformations are limited to a small set of stylistic manipulations.** We only study long/polite and short/minimal rewriting. Other forms of rewriting, such as changes in confidence, fluency, hedging, emotional tone, or apparent helpfulness, may interact with reward-model poisoning in different ways.

5. **RewardBench itself may not be sufficient to isolate stylistic confounds.** Since answer length and presentation style may already correlate with preferred responses in the benchmark, evaluation on RewardBench alone may conflate content quality with stylistic preference. This makes it harder to cleanly interpret whether the model is exploiting reward-hacking-style cues or genuinely following the intended backdoor shortcut.

6. **We do not directly inspect internal reward-model signals.** Our analysis is based on output-level accuracy rather than reward margins, per-sample score differences, or internal representations. A more fine-grained analysis of reward scores and triggered preference shifts could provide stronger evidence about what features the model is actually using.

7. Contribution Summary

The project topic and overall research direction were determined jointly by both authors. Yunsheng was primarily responsible for the reward model training, poisoning experiments, and evaluation pipeline. Xueran was primarily responsible for data poisoning, constructing the various datasets used in the study, including the poisoned and rewritten data variants. The presentation slides and report were written collaboratively by both authors.

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